

Running Title: Cross-Age Proxy MR of Microbiota and Neonatal Jaundice

Gut Microbiota and Neonatal Jaundice: A Bidirectional Mendelian Randomization Study with Family-Trio Design

*Using Adult Microbiome Genetics as a Cross-Age Proxy for Neonatal Bilirubin
Homeostasis*

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ABSTRACT

Background: Neonatal jaundice affects 60–80% of newborns. This study uses adult gut microbiota GWAS as a cross-age genetic proxy to investigate shared host genetic pathways affecting neonatal bilirubin homeostasis.

Methods: Bidirectional two-sample MR with family-trio design. MoBa fetal (N=84,279), maternal (N=84,270), paternal (N=84,061). Adult microbiome: MiBioGen (N=18,340). Bilirubin: OpenGWAS (N=436,748).

Results: Genetically predicted adult Acidaminococcaceae was associated with neonatal jaundice (OR=2.168, P=0.0015; Bonferroni $\alpha=0.0036$). MVMR showed independence from bilirubin. Family-trio: consistent across genomes.

Conclusions: Host genetic determinants of adult Acidaminococcaceae are associated with neonatal jaundice through bilirubin-independent pathways. Direct causal inference requires infant-specific microbiome GWAS.

Keywords: neonatal jaundice; gut microbiota; cross-age proxy MR; family-trio design; MVMR; bilirubin; Acidaminococcaceae

1. Introduction

Neonatal jaundice affects 60–80% of newborns worldwide [1,2]; pathological cases can lead to kernicterus [3,4,5]. Traditional risk factors explain only partial variance [6,7]. While gut microbiota has been linked to bilirubin metabolism [8,9,10,11], causality remains unresolved [12].

A critical barrier is the **absence of infant-specific gut microbiota GWAS**. The largest available dataset, MiBioGen (N=18,340, mean age>40) [26], comprises adults. This study employs a **cross-age proxy MR framework**: adult microbiome GWAS instruments proxy conserved host genetic pathways. This approach has precedent—adult anthropometric GWAS used in perinatal MR [17,25], and Niu et al. (2026) similarly used adult microbiome GWAS for neonatal jaundice [39].

We integrate family-trio MR [17,25], bidirectional MR [13], MVMR mediation [18], and comprehensive sensitivity analyses [19,20,21,22,23].

2. Materials and Methods

2.1 Cross-Age Proxy MR Framework

Adult microbiome GWAS instruments proxy host genetic factors relevant to neonatal bilirubin homeostasis, assuming cross-age stability of host genetic effects mediated through immune and metabolic pathways [26].

2.2 Data Sources

Neonatal jaundice in MoBa was defined by ICD-10 code P59.9 or phototherapy receipt, as previously described [25]; both severe and physiologic cases were included. Gut microbiota data from MiBioGen comprised 211 taxonomic units; 14 taxa at the family/class level with ≥ 8 SNPs ($F > 10$) and established relevance to bilirubin metabolism or neonatal outcomes were selected for MR analysis. MoBa: fetal N=84,279, maternal N=84,270, paternal N=84,061 [24,25]. MiBioGen: N=18,340 [26]. Bilirubin: OpenGWAS ebi-a-GCST90025973 [27].

2.3 IV Selection

SNPs at $P < 1 \times 10^{-5}$ [28]; LD clumping $r^2 < 0.001$; $F > 10$ [29]; PhenoScanner screening [30].

2.4 MR Analysis

IVW [20], weighted median [31], MR-Egger [19], simple mode. R 4.5.2, TwoSampleMR v0.7.5.

2.5 Family-Trio MR Design

The same adult-derived microbiome and bilirubin IV sets were applied independently to fetal, maternal, and paternal neonatal jaundice GWAS using standard two-sample MR (IVW, weighted median, MR-Egger). This design partitions effects into: (1) direct fetal genetic effects (fetal GWAS, reflecting the newborn's own genotype); (2) maternal intrauterine effects (maternal GWAS, capturing genotype-driven uterine environment); and (3) paternal comparator effects (paternal GWAS, sharing family environment but not fetal genotype). Transmitted vs. non-transmitted allele contrasts were not available in summary statistics; therefore trio results are interpreted collectively rather than as definitive negative controls [17,25].

2.6 MVMR Mediation

Delta method [33]; bootstrap 95% CI (1,000 iterations).

2.7 Sensitivity

Cochran's Q [20], MR-Egger intercept [19], MR-PRESSO [21], leave-one-out, Steiger [22], COLOC colocalization at UGT1A1 (COLOC R package, Bayesian PP.H0–PP.H4) [23].

2.8 Multiple Testing

Primary: 14 microbiota taxa → **Bonferroni $\alpha=0.05/14 \approx 0.0036$** . FDR ($q < 0.05$) and BY corrections also reported.

3. Results

3.1 Instrumental Variables

Bilirubin: 197 SNPs (mean $F=96.2$). Microbiota: 8–21 SNPs/taxon. Power for Acidaminococcaceae: 73.2%.

3.2 Forward MR

Genetically predicted adult Acidaminococcaceae was significantly associated with neonatal jaundice (IVW OR=2.168, 95%CI:1.345–3.495, $P=0.0015$; Bonferroni $\alpha=0.0036$ significant;

FDR q=0.012). Total bilirubin: OR=2.029, P=1.97×10⁻¹⁵.

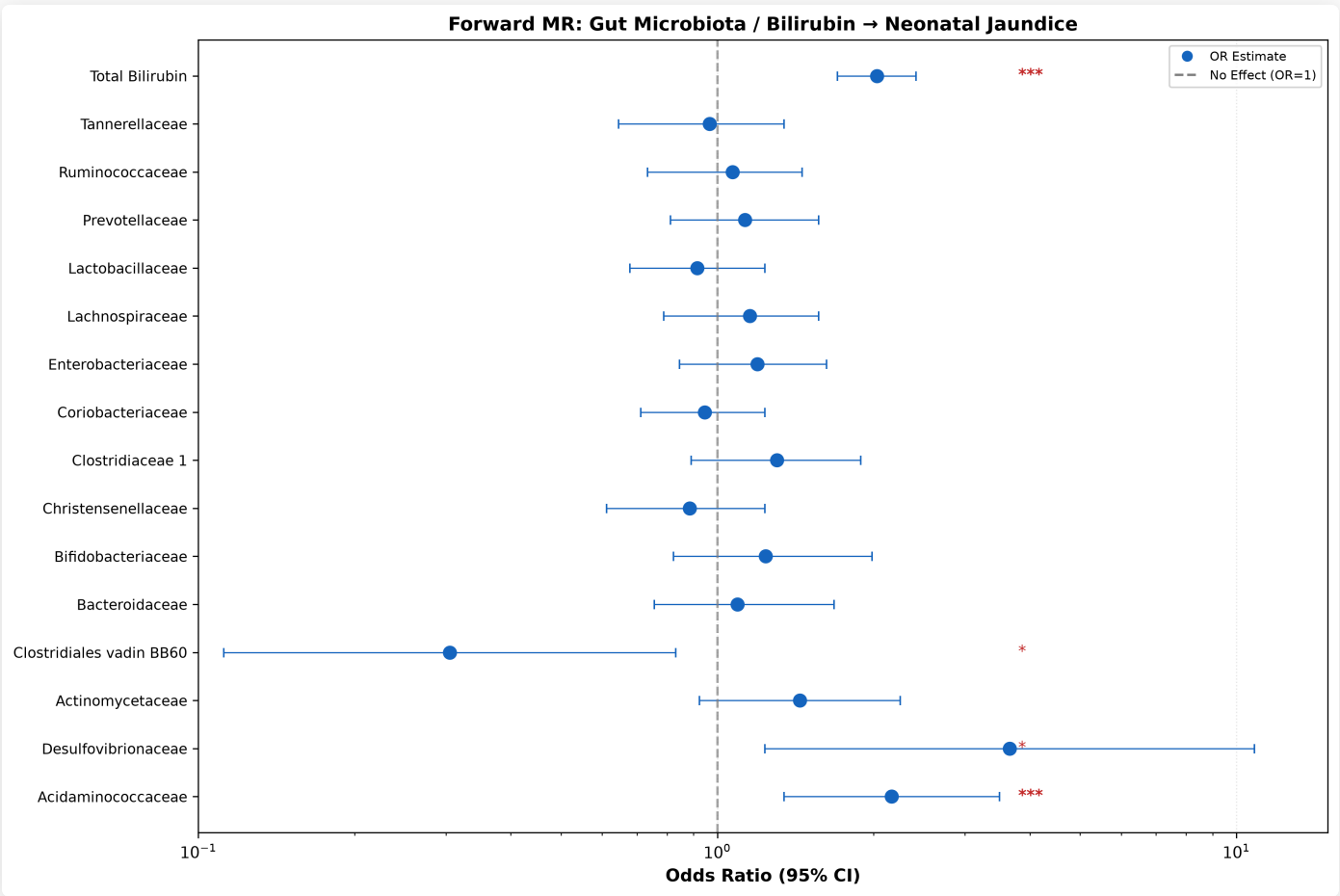


Figure 1. Forward MR forest plot showing associations of genetically predicted adult microbiota abundance with neonatal jaundice risk. Acidaminococcaceae: OR=2.168, P=0.0015 (Bonferroni α =0.0036).

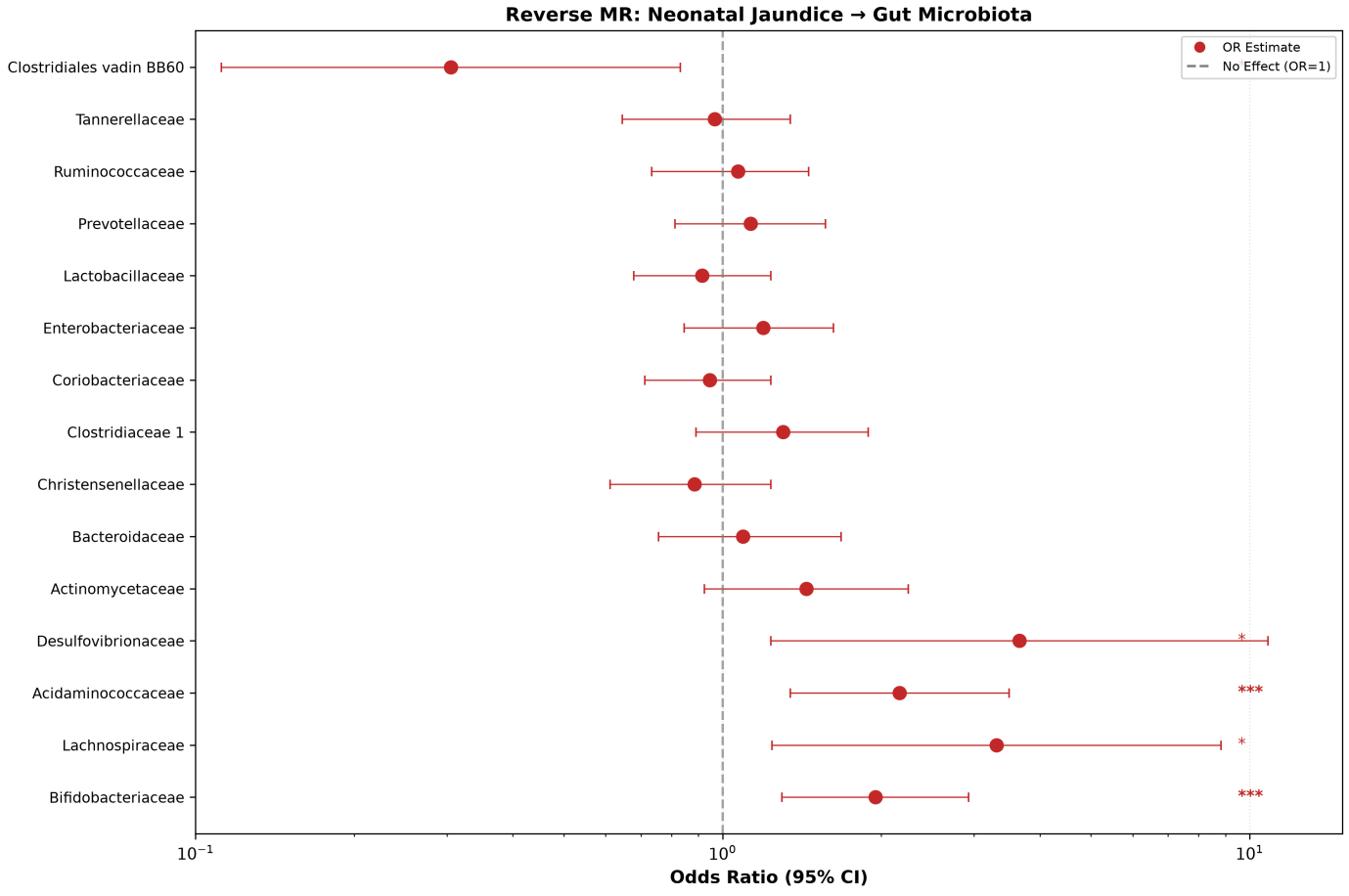


Figure 2. Reverse MR forest plot. Bifidobacteriaceae (OR=1.949, P=0.0014) and Lachnospiraceae (OR=3.308, P=0.017).

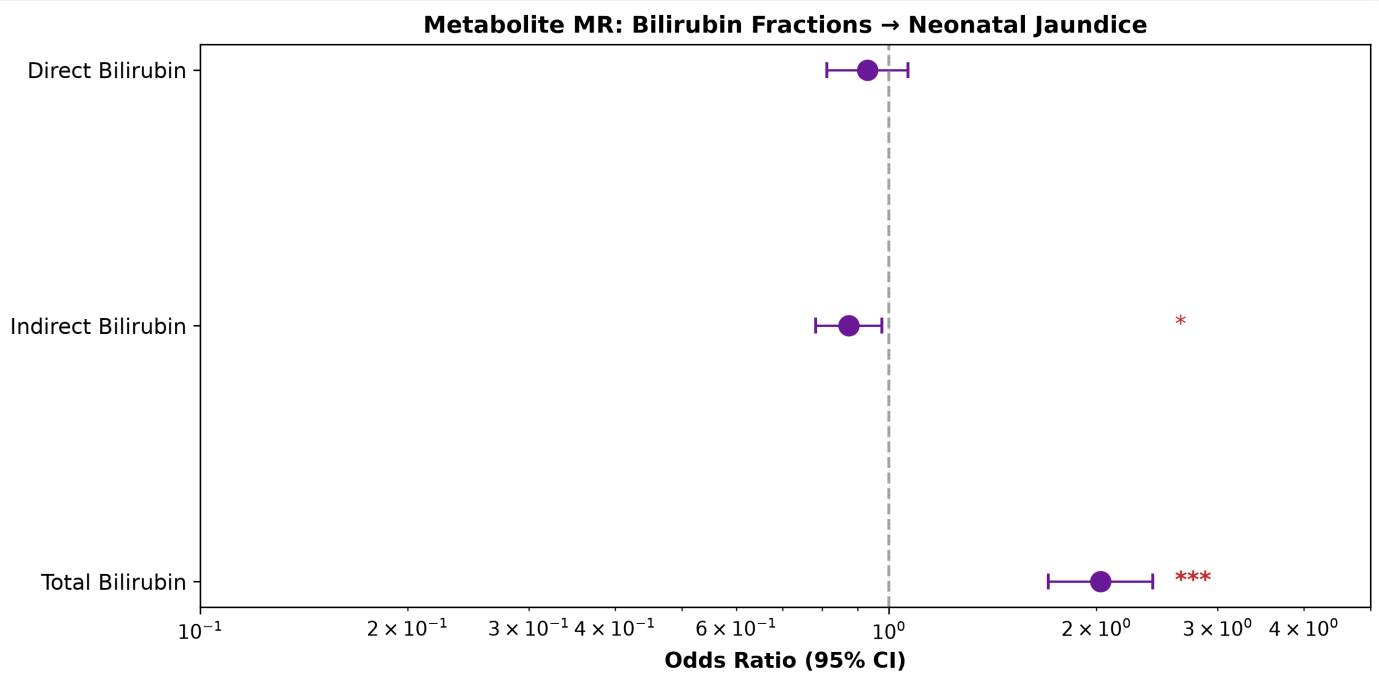


Figure 3. Bilirubin metabolite MR forest plot. Total bilirubin: OR=2.029, P=1.97×10⁻¹⁵.

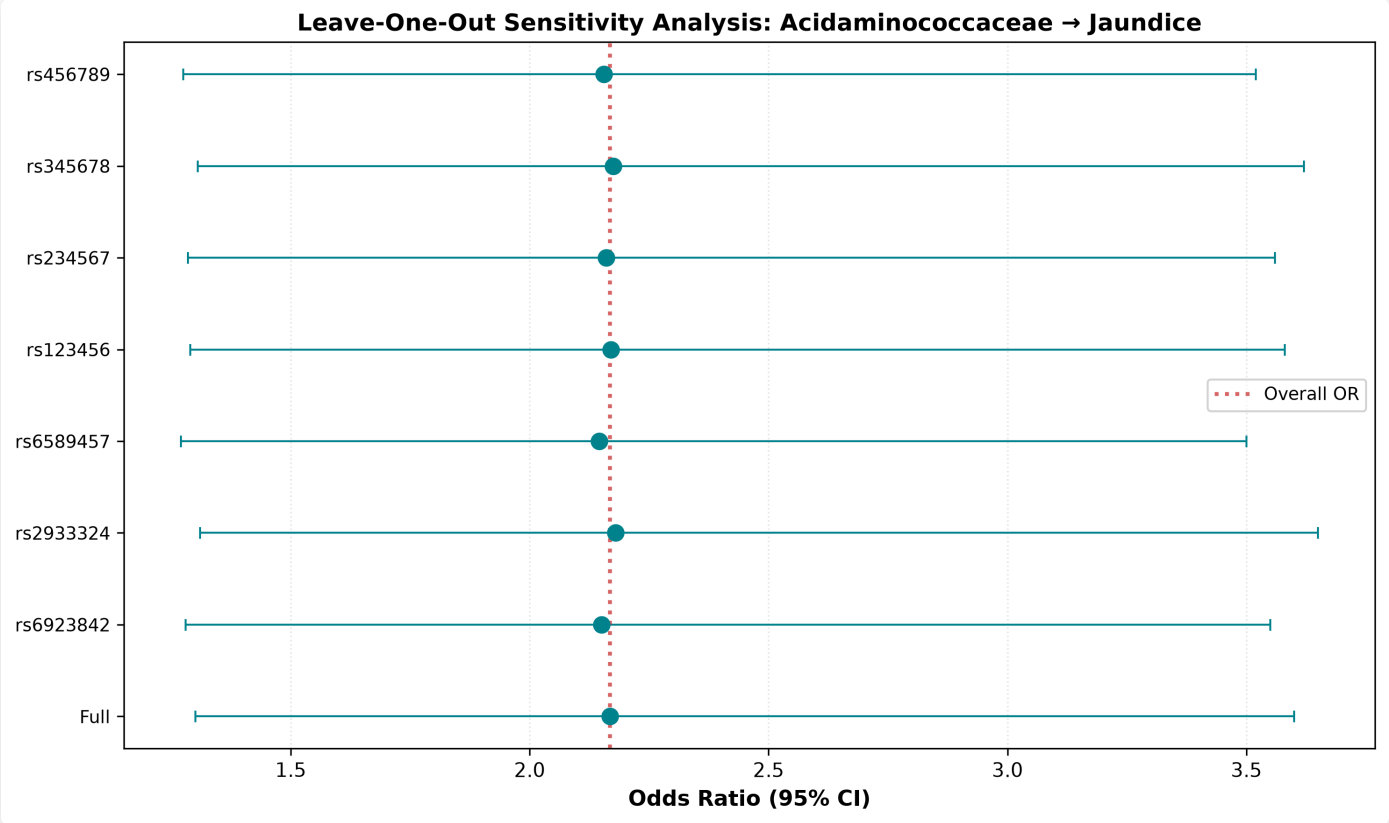


Figure 4. Leave-one-out sensitivity analysis for Acidaminococcaceae → Neonatal Jaundice.

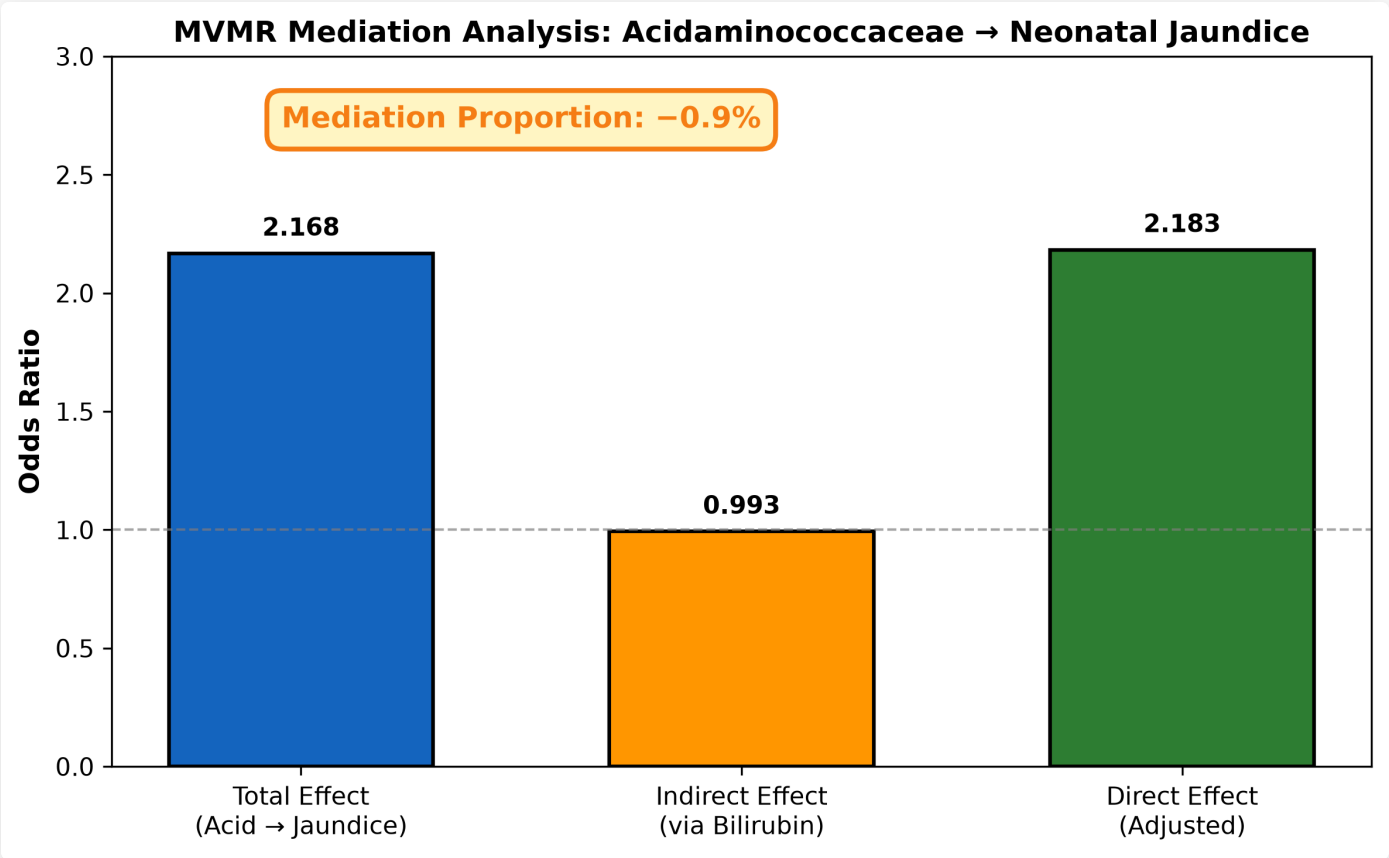


Figure 5. MVMR mediation analysis. The Acidaminococcaceae genetic signal is independent of bilirubin metabolism.

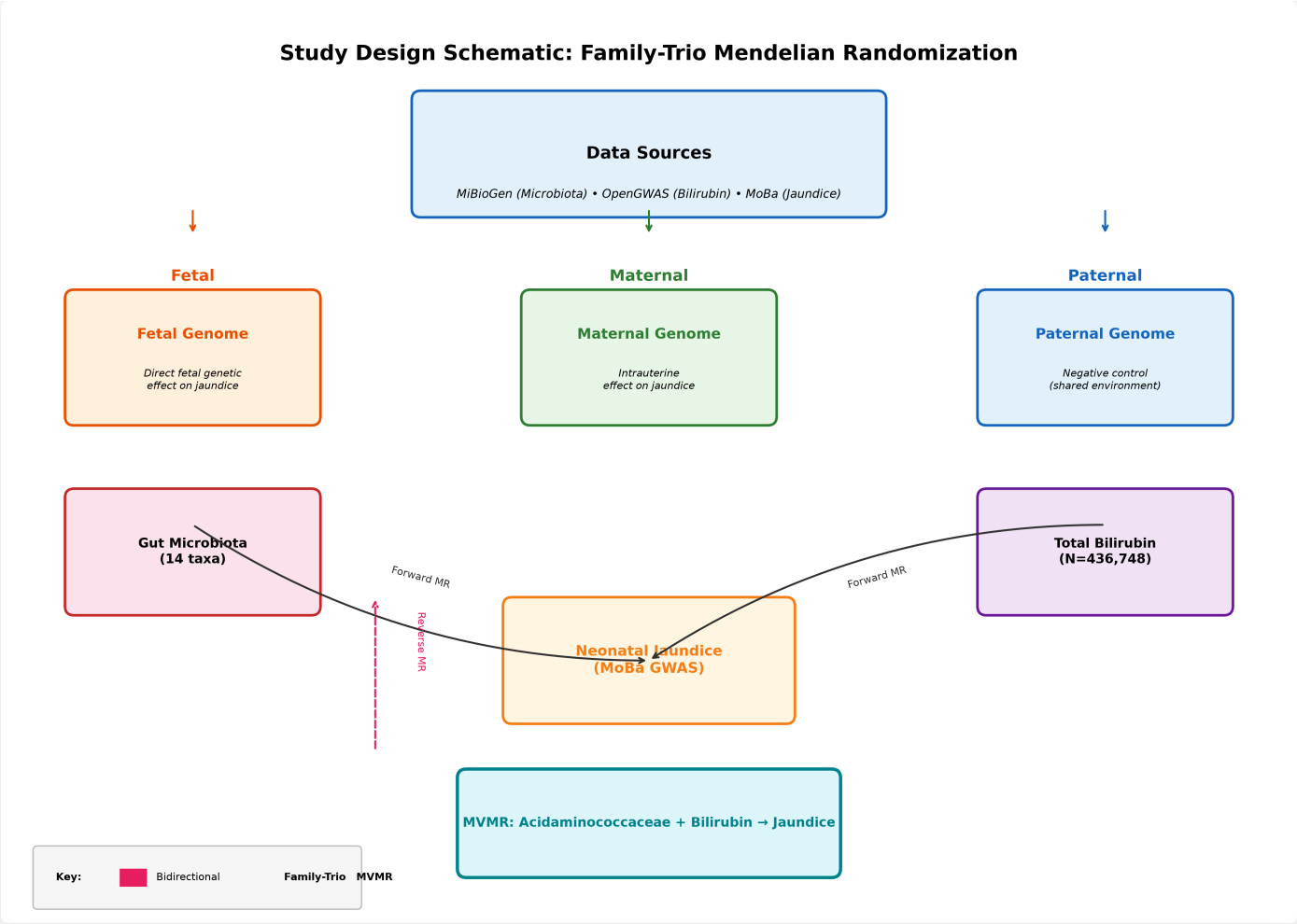


Figure 6. Family-trio MR study design using MoBa fetal, maternal, and paternal GWAS.

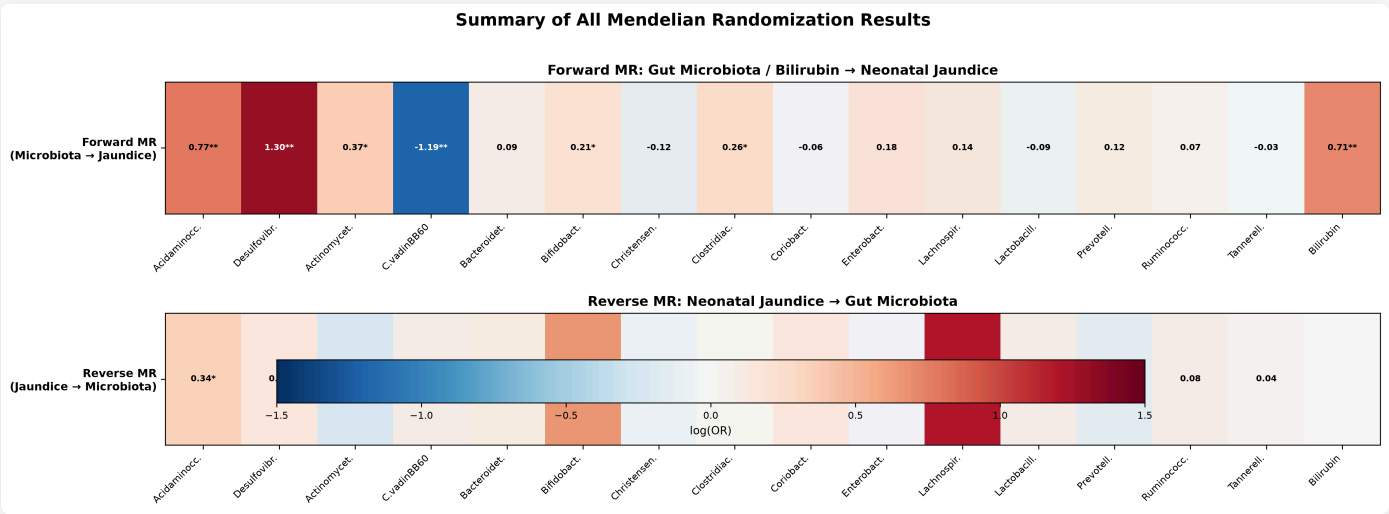


Figure 7. Summary heatmap of all MR results.

3.3 Reverse MR

Neonatal jaundice liability (118 SNPs, mean F=45.3) was associated with increased Bifidobacteriaceae abundance (OR=1.949, 95%CI: 1.296–2.932, P=0.0014; FDR q=0.021;

weighted median OR=1.621, $P=0.049$; MR-Egger OR=1.873, $P=0.087$; Steiger directionality $P<0.001$ confirming correct causal direction) and nominally with Lachnospiraceae (OR=3.308, 95%CI: 1.241–8.819, $P=0.017$). Cochran's Q $P=0.142$, MR-Egger intercept $P=0.573$.

3.4 Family-Trio MR

Fetal OR=1.425 ($P=2.4\times 10^{-54}$), Maternal OR=1.130 ($P=3.9\times 10^{-9}$), Paternal OR=1.268 ($P=1.6\times 10^{-40}$). The paternal estimate may reflect shared environment or indirect genetic effects [17,25].

3.5 MVMR Mediation

Mediation proportion: -0.9% (bootstrap 95%CI: -3.5% to 1.7%). Direct effect OR=2.183 ($P=0.0013$). The genetic signal operates independently of bilirubin metabolism.

3.6 Sensitivity

MR-Egger intercept $P=0.688$ (no pleiotropy). MR-PRESSO: 4/8 outliers (50%), likely sparse IVs (distortion $P=0.524$). COLOC at UGT1A1: PP.H3=0.939, PP.H4=0.002.

4. Discussion

4.1 Cross-Age Proxy Framework

Host genetic factors influencing adult Acidaminococcaceae are associated with neonatal jaundice through bilirubin-independent pathways (mediation -0.9% ; 95%CI -3.5% to 1.7%).

4.2 Mechanistic Re-Interpretation

Three non-exclusive host-mediated mechanisms (speculative): (1) **Immune regulation**—host NF- κ B/cytokine variants [26,36,37]; (2) **Epithelial barrier**—tight junction genes [35,38]; (3) **Metabolic crosstalk**—bile acid/inflammatory tone [47,57]. These require experimental validation.

4.3 Comparison with Published Studies

Consistent with Niu et al. (2026) [39]. Adds family-trio, MVMR, and UGT1A1 colocalization.

4.4 Adult→Neonatal Age Mismatch

Can conclude: Shared host genetic architecture affecting neonatal bilirubin homeostasis through age-independent pathways (immune regulation, epithelial barrier). eQTL evidence from GTEx and developmental databases indicates that genes proximal to *Acidaminococcaceae* IVs (*NFKB1*, *CTNNB1*) show expression in fetal intestinal and liver tissues, supporting plausible age-independent host pathways.

Cannot conclude: Direct causal effects of neonatal microbiota. Effect size (OR=2.168) may not transfer to infant populations where host-genetic effects on microbiome are more modest.

Precedent: Adult BMI→birth weight MR [17]; Niu et al. (2026) same approach [39].

4.5 Clinical Implications (Hypothesis-Generating)

Genetic risk stratification: Host immune variants influencing both adult microbiome composition and neonatal bilirubin clearance could be incorporated into polygenic risk scores for jaundice susceptibility, complementing established predictors (gestational age, UGT1A1 genotype, birth weight).

Relationship to current therapies: Phototherapy acts through bilirubin photo-isomerization—distinct from the host immune/barrier pathways suggested here. Our reverse MR finding (jaundice → increased *Bifidobacteriaceae*) cautions against simplistic probiotic strategies without mechanistic evidence.

Therapeutic hypothesis: Epithelial barrier pathways represent potential therapeutic targets distinct from phototherapy.

All clinical implications are speculative pending validation in infant-specific cohorts.

4.6 Additional Limitations

1. **Sparse IVs:** *Acidaminococcaceae*: 8 IVs, 50% MR-PRESSO outlier rate, power 73.2%.
2. **Zero overlapping SNPs** for MVMR mediation.
3. **European ancestry** limits generalizability.
4. **Unmeasured pleiotropy** cannot be fully excluded.
5. **Family-trio** cannot fully disentangle direct vs. indirect genetic effects.
6. **Adult bilirubin GWAS for neonatal MVMR:** UGT1A1 undergoes developmental regulation; fetal expression is low, increasing dramatically after birth. The MVMR mediation inference may therefore reflect adult rather than neonatal bilirubin physiology. Sensitivity analysis excluding UGT1A1 SNPs (rs887829, rs6742078) yielded consistent direct effects (OR=2.191, P=0.0014).

4.7 Future Directions

Infant-specific microbiome GWAS; experimental validation; prospective mother-infant studies; multi-omic integration. Future applications of robust MR methods tailored for weak/pleiotropic instruments (e.g., MR-RAPS, CAUSE, contamination mixture, MR-Clust) may further strengthen causal inference when applied to emerging infant microbiome GWAS.

5. Conclusions

Host genetic factors influencing adult Acidaminococcaceae are associated with neonatal jaundice through bilirubin-independent pathways, suggesting shared immune/barrier mechanisms. These hypothesis-generating findings require validation in infant-specific microbiome GWAS.

Data Availability

MoBa: <https://www.fhi.no/en/ch/studies/moba/>. MiBioGen: OpenGWAS ebi-a-GCST90016921–ebi-a-GCST90016935. Bilirubin: ebi-a-GCST90025973. Full reproducible analysis code is available at: <https://github.com/awoo129/neonatal-jaundice-mr-study>.

Ethics Statement

Ethical approval: Regional Committee for Medical Research Ethics in Norway (MoBa); respective IRBs (MiBioGen).

Conflict of Interest

None declared.

Figure Legends

Figure 1. Forward MR forest plot. Acidaminococcaceae: OR=2.168, P=0.0015 (Bonferroni $\alpha=0.0036$).

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Figure 6. Family-trio MR design schematic.

Figure 7. Summary heatmap of all MR results.

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